

VITTO ASSESSMENT

Frontend :- [React.js](#) (Vite)

Backend :- [Node.js](#)

Database :- MySQL

Frontend

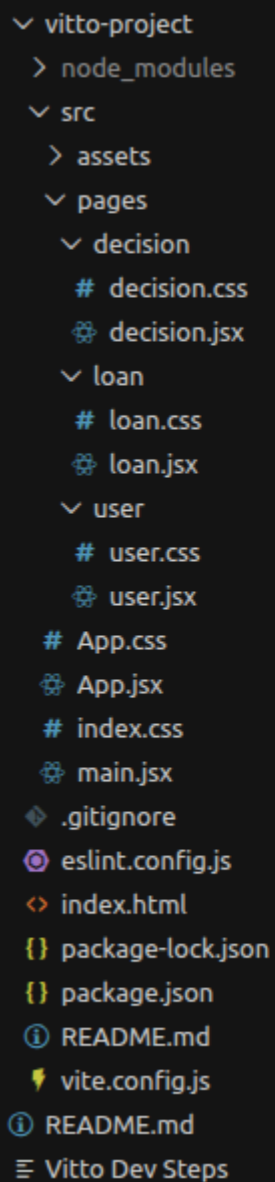
Front End is done using Reacts Vite Version :- **create-vite@9.0.4**

```
npm create vite@latest vitto-project -- --template react
```

npm needed to install to get modules :-

```
npm install react-router-dom
```

```
npm init
```



The screenshot shows a file explorer view of a project named 'vitto-project'. The structure is as follows:

- vitto-project
 - node_modules
 - src
 - assets
 - pages
 - decision
 - decision.css
 - decision.jsx
 - loan
 - loan.css
 - loan.jsx
 - user
 - user.css
 - user.jsx
 - App.css
 - App.jsx
 - index.css
 - main.jsx
 - .gitignore
 - eslint.config.js
 - index.html
 - package-lock.json
 - package.json
 - README.md
 - vite.config.js
 - README.md
 - Vitto Dev Steps

Above is the structure of the Frontend where functionalities are divided folder wise

src > pages - These Show the pages which are to going to be seen on the website

Inside this there is decision folder , loan folder and user folder

1) Decision

Here the Final decision after calculation the Users monthly income and the loan he want is calculated and the final report i.e if the Loan is approved or not is shown

Code :-

```
import React from 'react';
import { useLocation, useNavigate } from 'react-router-dom';

function Decision() {
  const { state } = useLocation();
  const navigate = useNavigate();

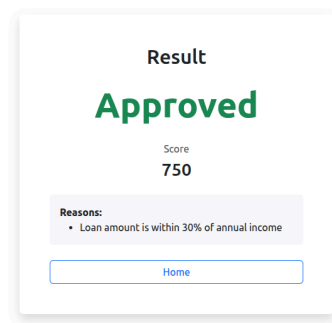
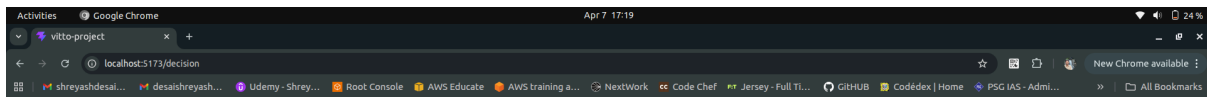
  if (!state) return <div className="text-center mt-5">No Data</div>;

  return (
    <div className="container mt-5 p-5 shadow rounded bg-white text-center"
      style={{ maxWidth: '500px' }}>
      <h2 className="mb-4">Result</h2>
      <h1 className={`display-4 fw-bold ${state.decision === 'Approved'
        ? 'text-success' : 'text-danger'}`}>
        {state.decision}
      </h1>
      <div className="my-4">
        <p className="text-muted mb-1">Score</p>
        <h3>{state.score}</h3>
      </div>
      <div className="text-start mb-4 bg-light p-3 rounded">
        <strong>Reasons:</strong>
        <ul className="mb-0">
          {state.reasons.map((r, i) => <li key={i}>{r}</li>)}
        </ul>
      </div>
    </div>
  );
}
```

```

    <button onClick={() => navigate('/')} className="btn
btn-outline-primary w-100">Home</button>
  </div>
);
}
export default Decision;

```



2) Loan

Here after the user Inputs his personal details the amount required for loan , the tenure (in months) which he is going to pay and the loan purpose is entered.

After this the final decision are made if the loan is to be approved or not.

Code :-

```

import React, { useState } from 'react';
import { useNavigate } from 'react-router-dom';

function Loan() {
  const [loan, setLoan] = useState({ amount: '', purpose: '', tenure: '12' });
  const navigate = useNavigate();

  const handleApply = async (e) => {

```

```

e.preventDefault();
const userId = localStorage.getItem('userId');
const rev = localStorage.getItem('tempRev');

const dResp = await fetch('http://localhost:4000/decision', {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({ monthlyRevenue: rev, loanAmount:
loan.amount })
  });
const dRes = await dResp.json();

await fetch('http://localhost:4000/loan/apply', {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({ userId, ...loan, status:
dRes.data.decision, score: dRes.data.score })
  });

navigate('/decision', { state: dRes.data });
};

return (
  <div className="container mt-5 p-5 " style={{ maxWidth: '500px' }}>

    <h2 className="text-center mb-4">Loan Details</h2>

    <form onSubmit={handleApply}>

      <div className="form-floating mb-3">
        <input type="number" className="form-control" onChange={e =>
setLoan({...loan, amount: e.target.value})} required />
        <label>AMOUNT</label>
      </div>

      <div className="form-floating mb-3">
        <input type="text" className="form-control" onChange={e =>
setLoan({...loan, purpose: e.target.value})} required />
        <label>PURPOSE</label>
      </div>

      <select className="form-select mb-4" onChange={e =>
setLoan({...loan, tenure: e.target.value})}>

```

```
        <option value="12">12 Months</option>
        <option value="24">24 Months</option>
        <option value="36">36 Months</option>
    </select>

    <button className="btn btn-success w-100 py-2 fw-bold">SUBMIT
APPLICATION</button>

</form>

</div>
);
}
export default Loan;
```

Page View :-

3) User

The Basic first page when the website is opened is the user Page.

Here the Information of the Applicant who needs loan is entered. This contains the name , pan number , business type and monthly revenue of the person.

Code :-

```
import React, { useState } from 'react';
import { useNavigate } from 'react-router-dom';

function User() {
  const [data, setData] = useState({ name: '', pan: '', businessType: '', monthlyRevenue: '' });
  const navigate = useNavigate();

  const handleNext = async (e) => {
    e.preventDefault();
    const resp = await fetch('http://localhost:4000/register', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(data)
    });
    const res = await resp.json();
    if (res.status === 'success') {
      localStorage.setItem('userId', res.data.userId);
      localStorage.setItem('tempRev', data.monthlyRevenue);
      navigate('/loan');
    }
  };

  return (
    <div className="container mt-5 p-5 " style={{ maxWidth: '500px' }}>

      <h2 className="text-center mb-4">Business Profile</h2>

      <form onSubmit={handleNext}>

        <div className="form-floating mb-3">
          <input type="text" className="form-control" onChange={e =>
            setData({...data, name: e.target.value})} required />
          <label>NAME</label>
        </div>


```



```

    <div className="form-floating mb-3">
      <input type="text" className="form-control" onChange={e =>
setData({...data, pan: e.target.value})} required />
      <label>PAN</label>
    </div>

    <select className="form-select mb-3" onChange={e =>
setData({...data, businessType: e.target.value})} required>
      <option value="">Business Type</option>
      <option value="Retail">Retail</option>
      <option value="Manufacturing">Manufacturing</option>
      <option value="Services">Services</option>
    </select>

    <div className="form-floating mb-3">
      <input type="number" className="form-control" onChange={e =>
setData({...data, monthlyRevenue: e.target.value})} required />
      <label>MONTHLY REVENUE</label>
    </div>

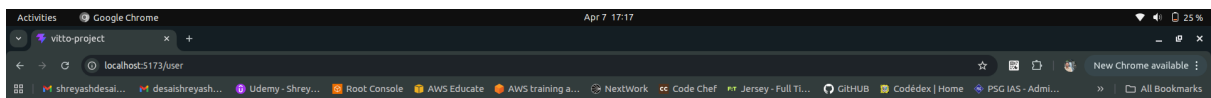
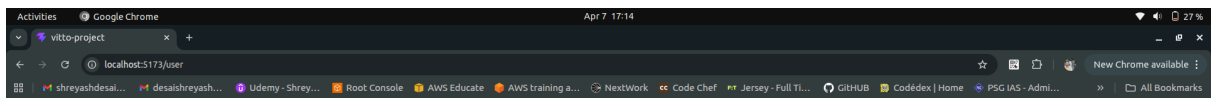
    <button className="btn btn-primary w-100 py-2
fw-bold">NEXT</button>

  </form>

</div>
);
}
export default User;

```

Page View :-



Business Profile

NAME	Samarjit
PAN	879456123
Business Type	Retail
MONTHLY REVENUE	88004
NEXT	

Database :-

```
mysql> SELECT * from User;
+----+-----+-----+-----+-----+-----+
| id | name   | pan      | business_type | monthly_revenue | created_at          |
+----+-----+-----+-----+-----+-----+
| 1  | Shreyash | 1234354sdf | Retail        | 100000.00        | 2026-04-07 16:48:36 |
| 2  | Samarjit | 879456123 | Retail        | 88004.00         | 2026-04-07 17:17:30 |
+----+-----+-----+-----+-----+-----+
```

App.jsx :-

The Main Component which is only on the Main.jsx having Routes Route to navigate through Pages.

Code :-

```
import { useState } from 'react'
import User from './pages/user/user'
import Loan from './pages/loan/loan'
import Decision from './pages/decision/decision'
import { Navigate, Routes , Route } from 'react-router-dom'
import './App.css'

function App() {

  return (
    <>
      <Routes>
        <Route
          path="/"
          element={<Navigate to="/user" />}
        />
        <Route
          path='user'
          element={<User/>}
        />

        <Route
          path='loan'
          element={<Loan/>}
        />

        <Route
          path='decision'
          element={<Decision/>}
        />

      </Routes>
    </>
  )
}

export default App
```

Main.jsx :-

The most Important jsx in React App **main.jsx** . Basically the main point to load and the components only render inside this so no Full page reload instead just a component reload.
Concept Known as **Virtual DOM (VDOM)**

Code :-

```
import { BrowserRouter } from 'react-router-dom'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  <BrowserRouter>
    <App />
  </BrowserRouter>
)
```

Full Frontend Side Architecture :-

Database

This is The Database Structure for the Project :-

2 main tables **User** and **Loans**.

Database name **vitto_data**

```
mysql> show tables;
+-----+
| Tables_in_vitto_data |
+-----+
| Loans |
| User |
+-----+
2 rows in set (0.01 sec)

mysql> desc Loans;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| id | int | NO | PRI | NULL | auto_increment |
| user_id | int | NO | MUL | NULL | |
| amount | decimal(15,2) | NO | | NULL | |
| tenure | int | NO | | NULL | |
| purpose | text | NO | | NULL | |
| status | enum('Approved','Rejected','Pending') | YES | | Pending | |
| credit_score | int | YES | | NULL | |
| applied_at | timestamp | YES | | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+
8 rows in set (0.00 sec)

mysql> desc User;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| id | int | NO | PRI | NULL | auto_increment |
| name | varchar(255) | NO | | NULL | |
| pan | varchar(10) | NO | | NULL | |
| business_type | enum('Retail','Manufacturing','Services','Others') | NO | | NULL | |
| monthly_revenue | decimal(15,2) | NO | | NULL | |
| created_at | timestamp | YES | | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+
6 rows in set (0.00 sec)

mysql>
```

Database And Table Creation :-

```
CREATE DATABASE IF NOT EXISTS vitto_data;
USE vitto_data;

CREATE TABLE IF NOT EXISTS User (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(255) NOT NULL,
    pan VARCHAR(10) NOT NULL,
    business_type ENUM('Retail', 'Manufacturing', 'Services', 'Others')
NOT NULL,
    monthly_revenue DECIMAL(15, 2) NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE IF NOT EXISTS Loans (
    id INT AUTO_INCREMENT PRIMARY KEY,
```

```
user_id INT NOT NULL,  
amount DECIMAL(15, 2) NOT NULL,  
tenure INT NOT NULL,  
purpose TEXT NOT NULL,  
status ENUM('Approved', 'Rejected', 'Pending') DEFAULT 'Pending',  
credit_score INT,  
applied_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
FOREIGN KEY (user_id) REFERENCES User(id) ON DELETE CASCADE  
);
```

Backend

This is the Backend made using [Node.js](#) mainly having 3 API /user , /loan, /decision

npm needed to install to get modules.

```
npm install express
```

```
npm install --save mysql2
```

```
npm install cors
```

1) /user

The 1st /user API is for the User information. This is having a Post method because we want to store info of user in the Database

Code :-

```
const express = require('express');
const pool = require('../db/db');
const { createResult } = require('../util/result');
const router = express.Router();

router.post('/register', (req, res) => {
  const { name, pan, businessType, monthlyRevenue } = req.body;
  const query = `INSERT INTO User (name, pan, business_type,
monthly_revenue) VALUES (?, ?, ?, ?)`;
  pool.execute(query, [name, pan, businessType, monthlyRevenue], (err,
data) => {
    res.send(createResult(err, data ? { userId: data.insertId } :
null));
  });
});

module.exports = router;
```

2) /loan

This is the /loan 2nd API for the loan information entering purpose. This is also a Post method

```
const express = require('express');
const pool = require('../db/db');
const { createResult } = require('../util/result');
const router = express.Router();

router.post('/apply', (req, res) => {
  const { userId, amount, tenure, purpose, status, score } = req.body;
  const query = `INSERT INTO Loans (user_id, amount, tenure, purpose, status, credit_score) VALUES (?, ?, ?, ?, ?, ?)`;
  pool.execute(query, [userId, amount, tenure, purpose, status, score], (err, data) => {
    res.send(createResult(err, data));
  });
});

module.exports = router;
```

3) /decision

The /decision is the 3rd and final API of the Application showing the final loan approval/rejection details. This is a Post method.

```
const express = require('express');
const { createResult } = require('../util/result');
const router = express.Router();

router.post('/', (req, res) => {
  const { monthlyRevenue, loanAmount } = req.body;
  const annualIncome = parseFloat(monthlyRevenue) * 12;
  const maxLoanLimit = annualIncome * 0.30;

  let score = 750;
  let decision = "Approved";
  let reasons = ["Loan amount is within 30% of annual income"];

  if (parseFloat(loanAmount) > maxLoanLimit) {
    score = 350;
    decision = "Rejected";
    reasons = [`Requested amount exceeds 30% annual income limit of ${maxLoanLimit.toFixed(0)}`];
  }
});
```



```
    res.send(createResult(null, { decision, score, reasons }));
  });
}

module.exports = router;
```

[server.js](#) :-

The main Backend heart of [Node.js](#) basically having the web url (localhost in this case). Also having the 3 API route connections which are imported from server folder containing the page use API's and using it with the app.use which is the variable extracted from the express imported to make the routing available.

app.use is used for the routing purpose and the cors main function to enable **Cross Origin Resource Sharing** without which the browser won't give permission to start our application.

Code :-

```
const express = require("express");
const cors = require('cors');
const userRouter = require('./route/user');
const loanRouter = require('./route/loan');
const decisionRouter = require('./route/decision');

const app = express();
app.use(cors());
app.use(express.json());

app.use('/', userRouter);
app.use('/loan', loanRouter);
app.use('/decision', decisionRouter);

app.listen(4000, () => {
  console.log("Server running on port 4000");
});
```

[db.js](#) :-

Having a pool by importing mysql2 module and in a constant named **pool** inserted the details of user, password, database name, and host.

This same module is exported to all API server to just use the ready made function rather than always configuring with the database for all individual pages.

Code :-

```
const mysql = require('mysql2')

const pool = mysql.createPool({
  host: 'localhost',
  user: 'root',
  password: 'root',
  database: 'vitto_data'
})

module.exports = pool
```

Full Backend Architecture :-

```
✓ VITTO
  ✓ server
    ✓ db
      JS db.js
    > node_modules
    ✓ route
      JS decision.js
      JS loan.js
      JS user.js
    ✓ util
      JS result.js
    {} package-lock.json
    {} package.json
    JS server.js
```